

Theme 2 – Lab Robotics

1.

Source	Key Ideas
Motsharg, E., Vunder, V., Raudmae, R., Muro, M., Drikkit, I., Tsigirinski, L., Koidam, R., Aabloo, A., & Kruusamae, K. (2024). Robotont 3-an accessible 3D-printable ROS-supported open-source mobile robot for education and research. <i>Frontiers in Robotics and AI</i> , 11, 1406645-. https://doi.org/10.3389/frobt.2024.1406645	The main ROS-based software system remains unchanged from the previous generation. Currently, the main software stack uses ROS 1 Noetic.
My Understanding	Synthesis
Robotont-3 is a successful research robot built using ROS architecture, using ROS1 Noetic, not using the latest ros2 based systems.	This creates the gap that most developers out of fear for unstable software, 2 years after the release of ROS2 still use ROS1. The Gap of encouragement for developers to migrate to ros2 architectures.

2.

Source	Key Ideas
Sun, X., & Zeng, Q. (2024). Research on Robot Operating System (ROS) Integration in Automated Logistics. <i>Applied Mathematics and Nonlinear Sciences</i> , 9(1). https://doi.org/10.2478/amns-2024-2388	ROS is peer to peer, Modular Supports multiple languages Free and open source

My Understanding	Synthesis
ROS is an important operating system for Robotics and is built upon Linux architecture. Incredibly modular due to the use of multiple distributed nodes which may or may not be part of the same programming languages.	This has allowed ROS to be the primary operating system when it comes to robotics projects, those including path planning, slam, etc.

3.

Source	Key Ideas
Ozturk, O. (2024). A Methodology for end user programming of ROS-based service robots using jigsaw metaphor and ontologies. <i>Intelligent Service Robotics</i> , 17(4), 745–757. https://doi.org/10.1007/s11370-024-00528-3	Use block-based software End users do not need to understand ROS Convert to Ros (pg. 750)
My Understanding	Synthesis

The Block-based software used has created a conversion system so that users can use normal software to execute ROS commands. The software will convert users' inputs into different ROS commands	This proves that ROS is more than just a ROBOTICS thing but also used in other software architectures. Common ROS projects like this still lack full autonomy and the author aims to reduce this gap.
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4.

Source	Key Ideas
Sun, Z., Divakar Nangeelil, K., & Searcy, H. (2024). Developing a remote gamma-ray spectra collection system (RGSCS) by coupling a high purity Germanium (HPGe) detector with a cosmicguard background reduction device. <i>HardwareX</i> , 17, e00513–e00513. https://doi.org/10.1016/j.ohx.2024.e00513	Used 7 separate DYNAMIXEL 5 degrees movements ROS compatibility
My Understanding	Synthesis

Smooth movement is produced using 7 separate DYNAMIXEL servos which allowed the arm joints to move freely and smoothly by just using abstracted python programs. It also has PID gains for helping with the smoothness of the movement.	This is the proof of successful use of the DYNAMIXEL motors with ROS and python using the DynamixelSDK, but the gap comes in controlling the specific models of XL330 servos.
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5.

Source	Key Ideas
Ramos, O., Casas, D., Cifuentes, C. A., & Jimenez, M. F. (2024). URHAND: Hand Prosthesis for Identifying Objects for Activities of Daily Living. <i>IEEE Transactions on Instrumentation and Measurement</i> , 1–1. https://doi.org/10.1109/TIM.2024.3470013	AI, Deep learning algorithms
My Understanding	Synthesis
MX 106 motors from DYNAMIXEL are so useful, they can output position, current as well as load on the motors. This means the data output can be used as validation filters for Machine learning models.	This tells one that the DYNAMIXEL servos are quite useful in a lot of ML stuff. But the gap remains in the fact that the specific model of XL 330, which the ADS-MT will be using.

6.

Source	Key Ideas
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Dow, C. (2024). <i>Internet of Things Programming Projects : Build Exciting IoT Projects Using Raspberry Pi 5, Raspberry Pi Pico, and Python</i> . (2nd ed.). Packt Publishing, Limited.	Chapter 11, The use of Ubuntu server on Pi4 model, installation of ROS2 system, ROS architectures. The ROS2 architecture was used for the IoT project.
My Understanding	Synthesis
Using Ros2 in a Raspberry Pi4 which usually comes with the installation of Ubuntu Server 22.0LTS, uses ros2 systems including ROS2 galactic, ROS2 humble.	This is using Ros2 for Ubuntu 22 with Pi4, Gap – We need to do research with Raspberry Pi 5 and Ros2 Jazzy, which can now be installed as Ubuntu 24.0LTS.

7.

Source	Key Ideas
Promkaew, N., Thammawiset, S., Srisan, P., Sanitchon, P., Tummawai, T., & Sukpancharoen, S. (2024). Development of metaheuristic algorithms for efficient path planning of autonomous mobile robots in indoor environments. <i>Results in Engineering</i> , 22, 102280-. https://doi.org/10.1016/j.rineng.2024.102280	Raspberry Pi 4 microcontroller as the main processing unit, working in conjunction with an Arduino Mega for controlling the DC motor drive through an MDD10A motor driver circuit. The robot is equipped with an RPLiDAR A1 sensor to read 360-degree distance values for mapping and obstacle avoidance.

My Understanding	Synthesis
Raspberry Pi 4 is an established model and is used again as a main controller in this case to control the Arduino Board which in turn modulates the DC motor drive to control motors.	This pointed the gap in research for Raspberry Pi5 to be used as a main controller for a simple robotics system with a motor that works with PWM.

Supporting non-peer reviewed references

8.

Open Robotics. (2024). *ROS 2 documentation: Humble*.
<https://docs.ros.org/en/humble/index.html>

9.

Open Robotics. (2024). *ROS 2 documentation: Jazzy*.
<https://docs.ros.org/en/jazzy/index.html>